

Michael Soprano, PhD

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Education

- Nov. 2019 – Jan. 2023 📖 **PhD in Computer Science, Mathematics and Physics**, Cum Laude. 35th cycle, University of Udine, Italy.
- Mar. 2015 – Mar. 2018 📖 **Master Degree in Computer Science**, 109/110, University of Udine, Italy.
- Oct. 2011 – Mar. 2015 📖 **Bachelor Degree in Web Technologies and Multimedia**, 101/110, University of Udine, Italy.
- Sep. 2006 – Jun. 2011 📖 **High School Diploma**, Head Technician Industrial Expert - Computer Science Specialization, 77/100, I.S.I.S. R. D'Aronco, Gemona del Friuli, Italy.

Visits

- Mar. 2022 – Jul. 2022 📖 **Visiting Student**, Centrum Wiskunde & Informatica, Human-Centered Data Analytics (HCDA) Group. Amsterdam, The Netherlands. Visiting period during PhD studies.

Work Experience

- Dec. 2023 - Today 📖 **Postdoctoral Researcher**. Department of Mathematics, Computer Science and Physics (DMIF), University of Udine, Italy.
PRIN 2022 Project – “MoT—The Measure of Truth: An Evaluation-Centered Machine-Human Hybrid Framework for Assessing Information Truthfulness” (Code No. 20227F2ZN3, CUP No. G53D23002800006), funded by the European Union – Next Generation EU – PNRR M4 C2 I1.1
Definition of an evaluation framework for misinformation detection systems, considering various aspects such as the creation of appropriately labeled datasets, addressing biases in dataset definition, and establishing suitable evaluation measures that specifically consider the concept of truthfulness in information.
- Mar. 2023 - Nov. 2023 📖 **Postdoctoral Researcher**. Department of Economic and Statistical Sciences (DIES), University of Udine, Italy.
Research project focused on exploring conversational agent-based eGovernment applications for the Italian public administration, with a particular emphasis on healthcare.
- Jan. 2022 - Mar. 2022 📖 **Self-employed Worker**. Polytechnic Department of Engineering and Architecture (DPIA), University of Udine, Italy.
Development of a web application dedicated to the comparability and reproducibility of research results in the field of combinatorial optimization.
- Jun. 2018 - Oct. 2019 📖 **Research Fellow**. Department of Mathematics, Computer Science and Physics (DMIF), University of Udine, Italy.
Research project co-funded by SISSA Medialab is dedicated to the design, implementation, and evaluation of an ecosystem that provides an alternative to the standard peer review activity in the scholarly publishing process.

Work Experience (continued)

- Jun. 2017 – Oct. 2019  **Self-employed Worker.** Datamantix S.r.l., Udine, Italy.
Design, development, and maintenance as a full-stack web developer of various types of web applications and a chatbot.
- Aug. 2017 – Feb. 2018  **Self-employed Worker.** Archeido S.r.l., Udine, Italy.
Development as a full-stack web developer of a web application for managing and visualizing financial assets.
- Nov. 2015 – Feb. 2016  **Self-employed Worker.** Troublezine.
Design, development, and maintenance as a full-stack web developer of a blog that offers news and events related to music.

Publications

- 1 Roitero, K., Wright, D., **Soprano, M.**, Augenstein, I., & Mizzaro, S. (2025). Collecting Cost-Effective, High-Quality Truthfulness Assessments with LLM Summarized Evidence. arXiv.
 doi:10.48550/arXiv.2501.18265. eprint: 2501.18265 (cs.IR)
- 2 Maddalena, E., Mizzaro, S., Roitero, K., Viviani, M., La Barbera, D., Modha, S., ... **Soprano, M.** (2024). Report on the 14th Italian Information Retrieval Workshop (IIR 2024). *SIGIR Forum*, 58(2), 1–13.
Retrieved from  <https://sigir.org/wp-content/uploads/2025/01/p15.pdf>
- 3 Spina, D., Roitero, K., Mizzaro, S., Della Mea, V., Da Ros, F., **Soprano, M.**, ... Tremamunno, A. (2024). Report on the Hands-On PhD Course on Responsible AI from the Lens of an Information Access Researcher. *SIGIR Forum*, 58(2), 1–61. Retrieved from
 <https://sigir.org/wp-content/uploads/2025/01/p07.pdf>
- 4 Roitero, K., **Soprano, M.**, La Barbera, D., Maddalena, E., & Mizzaro, S. (2024). Enhancing Fact-Checking: From Crowdsourced Validation to Integration with Large Language Models. In S. Mizzaro, E. Maddalena, M. Viviani, & K. Roitero (Eds.), *Proceedings of the 14th Italian Information Retrieval Workshop* (Vol. 3802, pp. 74–77). Udine, Italy: CEUR-WS.org. Retrieved from
 <https://ceur-ws.org/Vol-3802/paper13.pdf>
- 5 **Soprano, M.**, Roitero, K., Gadiraju, U., Maddalena, E., & Demartini, G. (2024). Longitudinal Loyalty: Understanding The Barriers To Running Longitudinal Studies On Crowdsourcing Platforms. *ACM Transactions on Social Computing*, 7(1–4). Journal Rank: Scimago (SJR) Q2 (2023).  doi:10.1145/3674884
- 6 Singh, J., **Soprano, M.**, Roitero, K., & Ceolin, D. (2024). Crowdsourcing Statement Classification to Enhance Information Quality Prediction. In M. Preuss, A. Leszkiewicz, J.-C. Boucher, O. Fridman, & L. Stampe (Eds.), *Proceedings of the 6th Multidisciplinary International Symposium on Disinformation in Open Online Media (MISDOOM 2024)* (pp. 70–85).  doi:10.1007/978-3-031-71210-4_5
- 7 La Barbera, D., Maddalena, E., **Soprano, M.**, Roitero, K., Demartini, G., Ceolin, D., ... Mizzaro, S. (2024). Crowdsourced Fact-checking: Does It Actually Work? *Information Processing & Management*, 61(5), 103792. Journal Ranks: Journal Citation Reports (JCR) Q1 (2023), Scimago (SJR) Q1 (2023).
 doi:10.1016/j.ipm.2024.103792
- 8 **Soprano, M.**, Roitero, K., La Barbera, D., Ceolin, D., Spina, D., Demartini, G., & Mizzaro, S. (2024). Cognitive Biases in Fact-Checking and Their Countermeasures: A Review. *Information Processing & Management*, 61(3), 103672. Journal Ranks: Journal Citation Reports (JCR) Q1 (2022), Scimago (SJR) Q1 (2022).  doi:10.1016/j.ipm.2024.103672
- 9 **Soprano, M.**, Roitero, K., Della Mea, V., & Mizzaro, S. (2023). Towards a Conversational-Based Agent for Health Services. In F. Falchi, F. Giannotti, A. Monreale, C. Boldrini, S. Rinzivillo, & S. Colantonio (Eds.), *Proceedings of the Italia Intelligenza Artificiale - Thematic Workshops co-located with the 3rd CINI National Lab AIIS Conference on Artificial Intelligence* (Vol. 3486, pp. 278–283). Pisa, Italy: CEUR-WS.org. Retrieved from  <https://ceur-ws.org/Vol-3486/96.pdf>

- 10 La Barbera, D., **Soprano, M.**, Roitero, K., Maddalena, E., & Mizzaro, S. (2023). Fact-Checking at Scale with Crowdsourcing: Experiments and Lessons Learned. In F. M. Nardini, N. Tonelotto, G. Faggioli, & A. Ferrara (Eds.), *Proceedings of the 13th Italian Information Retrieval Workshop* (Vol. 3448, pp. 85–90). Pisa, Italy: CEUR-WS.org. Retrieved from <https://ceur-ws.org/Vol-3448/paper-18.pdf>
- 11 Roitero, K., Barbera, D. L., **Soprano, M.**, Demartini, G., Mizzaro, S., & Sakai, T. (2023). How Many Crowd Workers Do I Need? On Statistical Power When Crowdsourcing Relevance Judgments. *ACM Transactions on Information Systems*. Journal Ranks: Journal Citation Reports (JCR) Q1 (2021), Scimago (SJR) Q1 (2021). [doi:10.1145/3597201](https://doi.org/10.1145/3597201)
- 12 Brand, E., Roitero, K., **Soprano, M.**, Rahimi, A., & Demartini, G. (2022). A Neural Model to Jointly Predict and Explain Truthfulness of Statements. *Journal of Data and Information Quality*, 15(1). Journal Ranks: Journal Citation Reports (JCR) Q3 (2021), Scimago (SJR) Q2 (2021). [doi:10.1145/3546917](https://doi.org/10.1145/3546917)
- 13 Ceolin, D., Primiero, G., **Soprano, M.**, & Wielemaker, J. (2022). Transparent Assessment of Information Quality of Online Reviews Using Formal Argumentation theory. *Information Systems*, 110, 102107. Journal Ranks: Journal Citation Reports (JCR) Q2 (2021), Scimago (SJR) Q1 (2021). [doi:10.1016/j.is.2022.102107](https://doi.org/10.1016/j.is.2022.102107)
- 14 Draws, T., La Barbera, D., **Soprano, M.**, Roitero, K., Ceolin, D., Checco, A., & Mizzaro, S. (2022). The Effects of Crowd Worker Biases in Fact-Checking Tasks. In *2022 ACM Conference on Fairness, Accountability, and Transparency* (pp. 2114–2124). [doi:10.1145/3531146.3534629](https://doi.org/10.1145/3531146.3534629)
- 15 **Soprano, M.**, Roitero, K., Bombassei De Bona, F., & Mizzaro, S. (2022). Crowd_Frame: A Simple and Complete Framework to Deploy Complex Crowdsourcing Tasks Off-the-Shelf. In *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining* (pp. 1605–1608). Conference Ranks: GGS A+, Core A*. [doi:10.1145/3488560.3502182](https://doi.org/10.1145/3488560.3502182)
- 16 **Soprano, M.**, Roitero, K., La Barbera, D., Ceolin, D., Spina, D., Mizzaro, S., & Demartini, G. (2021). The Many Dimensions of Truthfulness: Crowdsourcing Misinformation Assessments on a Multidimensional Scale. *Information Processing & Management*, 58(6), 102710. Journal Ranks: Journal Citation Reports (JCR) Q1 (2021), Scimago (SJR) Q1 (2021). [doi:10.1016/j.ipm.2021.102710](https://doi.org/10.1016/j.ipm.2021.102710)
- 17 Brand, E., Roitero, K., **Soprano, M.**, & Demartini, G. (2021). E-BART: Jointly Predicting and Explaining Truthfulness. In I. Augenstein, P. Papotti, & D. Wright (Eds.), *Proceedings of The 2021 Truth and Trust Online Conference* (pp. 18–27). Virtual Event. Retrieved from https://truthandtrustonline.com/wp-content/uploads/2021/10/TT02021_paper_16-1.pdf
- 18 Ceolin, D., Primiero, G., Wielemaker, J., & **Soprano, M.** (2021). Assessing the Quality of Online Reviews Using Formal Argumentation Theory. In M. Brambilla, R. Chbeir, F. Frasincar, & I. Manolescu (Eds.), *Web Engineering* (pp. 71–87). Conference Ranks (2021): GGS B-, Core B. [doi:10.1007/978-3-030-74296-6_6](https://doi.org/10.1007/978-3-030-74296-6_6)
- 19 Roitero, K., **Soprano, M.**, Portelli, B., De Luise, M., Spina, D., Mea, V. D., ... Demartini, G. (2021). Can The Crowd Judge Truthfulness? A Longitudinal Study On Recent Misinformation About COVID-19. *Personal and Ubiquitous Computing*, 27(1), 59–89. Journal Ranks: Journal Citation Reports (JCR) Q2 (2020), Scimago (SJR) Q1 (2021). [doi:10.1007/s00779-021-01604-6](https://doi.org/10.1007/s00779-021-01604-6)
- 20 Roitero, K., **Soprano, M.**, Portelli, B., Spina, D., Della Mea, V., Serra, G., ... Demartini, G. (2020). The COVID-19 Infodemic: Can the Crowd Judge Recent Misinformation Objectively? In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management* (pp. 1305–1314). Conference Ranks: GGS A+, Core A. [doi:10.1145/3340531.3412048](https://doi.org/10.1145/3340531.3412048)
- 21 Roitero, K., **Soprano, M.**, Fan, S., Spina, D., Mizzaro, S., & Demartini, G. (2020). Can The Crowd Identify Misinformation Objectively? The Effects of Judgment Scale and Assessor's Background. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 439–448). Conference Rank: GGS A++, Core A*. [doi:10.1145/3397271.3401112](https://doi.org/10.1145/3397271.3401112)

- 22 **Soprano, M.**, Roitero, K., & Mizzaro, S. (2019a). Bias and Fairness in Effectiveness Evaluation by Means of Network Analysis and Mixture Models. In Agosti, Maristella and Di Buccio, Emanuele and Melucci, Massimo and Mizzaro, Stefano and Pasi, Gabriella and Silvestri, Fabrizio (Ed.), *Proceedings of the 10th Italian Information Retrieval Workshop* (Vol. 2441, 6–7). Padova, Italy: CEUR-WS.org. Retrieved from <https://ceur-ws.org/Vol-2441/paper4.pdf>
- 23 **Soprano, M.**, Roitero, K., & Mizzaro, S. (2019b). HITS Hits Readersourcing: Validating Peer Review Alternatives Using Network Analysis. In Chandrasekaran, Muthu Kumar and Mayr, Philipp (Ed.), *Proceedings of the 4th Joint Workshop on Bibliometric-enhanced Information Retrieval and Natural Language Processing for Digital Libraries co-located with the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval* (Vol. 2414, 70–82). Paris, France: CEUR-WS.org. Retrieved from <https://ceur-ws.org/Vol-2414/paper7.pdf>
- 24 **Soprano, M.**, & Mizzaro, S. (2019a). Crowdsourcing Peer Review in the Digital Humanities? In *Book of Abstracts, 8th AIUCD Conference 2019 – Pedagogy, Teaching, and Research in the Age of Digital Humanities* (p. 251). Udine, Italy. Retrieved from http://aiucd2019.uniud.it/wp-content/uploads/2020/03/AIUCD2019-BoA_DEF.pdf
- 25 **Soprano, M.**, & Mizzaro, S. (2019b). Crowdsourcing Peer Review: As We May Do. In Manghi, Paolo and Candela, Leonardo and Silvello, Gianmaria (Ed.), *Digital Libraries: Supporting Open Science* (Vol. 988, 259–273). [doi:10.1007/978-3-030-11226-4_21](https://doi.org/10.1007/978-3-030-11226-4_21)
- 26 Roitero, K., **Soprano, M.**, Brunello, A., & Mizzaro, S. (2018). Reproduce and Improve: An Evolutionary Approach to Select a Few Good Topics for Information Retrieval Evaluation. *Journal of Data and Information Quality*, 10(3). Journal Rank: Scimago (SJR) Q2 (2018). [doi:10.1145/3239573](https://doi.org/10.1145/3239573)
- 27 Roitero, K., **Soprano, M.**, & Mizzaro, S. (2018). Effectiveness Evaluation with a Subset of Topics: A Practical Approach. In *The 41st International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 1145–1148). Conference Rank: GGS A++, Core A*. [doi:10.1145/3209978.3210108](https://doi.org/10.1145/3209978.3210108)

Metrics

	Scholar	Scopus	ACM Digital Library
Articles	27	20	16
Citations	356	187	100
Citations / Articles	13.18	9.35	6.00
h-index	10	7	–
i10-index	11	–	–

Talks

- 2024  **Deploying Complex Crowdsourcing Tasks Without Fear: An Introduction to Crowd_Frame.** *The 3rd Italian Conference on Big Data and Data Science*, September 17, 2024. Pisa, Italy. https://michaelsoprano.com/media/talks/An-Introduction-To-Crowd_Frame.pdf
- 2023  **Towards a Conversational-Based Agent for Healthcare.** *CARE: Conversational Agents and Responsible Ethics in Healthcare - Workshop at the European Federation of Medical Informatics Special Topic Conference 2023*, October 27, 2023. Turin, Italy. https://docs.google.com/presentation/d/1IZaI38oJ3FktP79dINPOL_9lqXmQf-a3TC7vf5qmfU/edit?usp=sharing
-  **Towards a Conversational-Based Agent for Health Services.** *Terzo Convegno Nazionale CINI sull'Intelligenza Artificiale*, May 29, 2023. Pisa, Italy. <https://www.ital-ia2023.it/submission/96/presentation>

Talks (continued)

- 2022  **Longitudinal Loyalty: Understanding the Barriers to Running Longitudinal Studies on Crowdsourcing Platforms.** *Centrum Wiskunde & Informatica*, June 16, 2022. Amsterdam, The Netherlands. https://docs.google.com/presentation/d/1YBHKOKY_TdaCfw09-pqm_jDGKYUiyXaKXj35pR3pSoM/edit?usp=sharing
-  **Crowd_Frame: A Simple and Complete Framework to Deploy Complex Crowdsourcing Tasks Off-the-Shelf.** *The 15th ACM International Conference on Web Search and Data Mining*, February 15th, Phoenix, U.S.A. Held remotely as a pre-recorded contribution due to the COVID-19 pandemic. <https://dl.acm.org/doi/abs/10.1145/3488560.3502182> (supplemental material section).
- 2020  **The COVID-19 Infodemic: Can the Crowd Judge Recent Misinformation Objectively?** *29th ACM International Conference on Information and Knowledge Management*, October 19th, 2020. Galway, Ireland. Held remotely as a pre-recorded contribution due to the COVID-19 pandemic. <https://dl.acm.org/doi/10.1145/3340531.3412048> (supplemental material section).
-  **Can The Crowd Identify Misinformation Objectively? The Effects of Judgment Scale and Assessor's Background** *43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, July 25th, 2020. Xi'an, China. Online, Held remotely as a pre-recorded contribution due to the COVID-19 pandemic. <https://dl.acm.org/doi/abs/10.1145/3397271.3401112> (supplemental material section).
- 2019  **Bias and Fairness in Effectiveness Evaluation by Means of Network Analysis and Mixture Model.** *Italian Information Retrieval Workshop*, September 16th, 2019. Padova, Italy. <https://docs.google.com/presentation/d/1F751FzwnxZjA0y0ctZy91BWEJzLCV19L9ff50Cn20Lo/edit?usp=sharing>

Teaching

- 2024/2025  **Web Applications.** Role: *Course Leader*. Master's Degree in "Management Engineering", University of Udine. I will be instructing the course in the first half of 2025. The course will have a total duration of 48 hours.
-  **Information Analysis and Processing for Training.** Role: *Course Leader*. Bachelor's Degree in "Sports Science", University of Udine. I designed and instructed the course, which consisted of 12 lectures with evaluation based on a written exam. The course had a total duration of 24 hours.
- 2020/2021–2024/2025  **Social Computing.** Role: *Teaching Assistant*. Instructor: Stefano Mizzaro. Bachelor's Degree in "Internet of Things, Big Data & Web", University of Udine. I prepared slides, delivered lectures, and designed the student projects for the laboratory component of the course. My contribution in class ranged from 4 to 12 hours per year.

Academic Activity

Organizational Roles

- 2025  21th Italian Research Conference on Digital Libraries (IRCDL 2025), Udine, February 20–21. *Website Chair*. <https://ircdl2025.uniud.it/>
- 2024  14th Italian Information Retrieval Workshop (IIR 2024), Udine, September 5–6. *Website Chair*. <https://iir2024.uniud.it/>
- 2023  CARE: Conversational Agents and Responsible Ethics in Healthcare - Workshop at the European Federation of Medical Informatics Special Topic Conference 2023 (EFMI STC 2023), Turin, October 27. *Organizer*.

Academic Activity (continued)

- 2022  Fourth Symposium on Biases in Human Computation and Crowdsourcing (BHCC 2022), Online, October 12-13. *Proceedings and Website Chair*. <https://bhcc2022.org/>.

Program Committee Roles

- 2024  12th AAAI Conference on Human Computation and Crowdsourcing (HCOMP 2024), Pittsburgh, Pennsylvania, USA. *Full Papers Track*.
-  47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2024), Washington D.C., USA. *Full and Short Papers Tracks*.
- 2023  45th European Conference on Information Retrieval (ECIR 2023), Dublin, Ireland. *Short Papers Track*.
- 2022  45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2022), Madrid, Spain. *Short Papers Track*.
- 2021  Third Symposium on Biases in Human Computation and Crowdsourcing (BHCC 2021), Online.

Reviewer

- 2025  Data Science and Engineering (DSEJ). <https://portal.issn.org/resource/ISSN/23641541>.
- 2023  Information Processing & Management (IP&M). <https://portal.issn.org/resource/ISSN/0306-4573>. Contribution registered on ORCID profile.
-  Springer Nature Scientific Report. <https://portal.issn.org/resource/ISSN/2045-2322>.
-  13th Italian Information Retrieval Workshop (IIR 2023), Pisa, Italy.
- 2022  Information Processing & Management. <https://portal.issn.org/resource/ISSN/0306-4573>.
-  Tenth AAAI Conference on Human Computation and Crowdsourcing (HCOMP 2022), Online.
-  Conference and Labs of the Evaluation Forum (CLEF 2022), Bologna, Italy.
-  44th European Conference on Information Retrieval (ECIR 2022), Stavanger, Norway.
-  Second Symposium on Biases in Human Computation and Crowdsourcing (BHCC 2021), Delft, Netherlands (Online).
- 2021  30th ACM International Conference On Information And Knowledge Management (CIKM 2021), Online.
-  32nd ACM Conference on Hypertext and Social Media, (HT 2021), Online.
- 2020  29th ACM International Conference On Information And Knowledge Management (CIKM 2020), Galway, Ireland (Online).
-  31st ACM Conference on Hypertext and Social Media (HT 2020), Online.
- 2019  15th Italian Research Conference On Digital Libraries (IRCDL 2019), Pisa, Italy.

Mentoring

- 2020 – Today  **Co-supervision** of the theses of 7 Master's (MSc) and 13 Bachelor's (BSc) students, as well as various advanced projects and seminars.

Affiliations

- 2021 – Today  **Member** of the Social, Mobile, Data & Crowd laboratory (SMDC), see: <https://smdc.uniud.it/smdc/en/members/>

Honors & Awards

- 2025  **PhD Award 2024**, Recipient of an award from the University of Udine for the best PhD thesis in the scientific area for 2023. Thesis title: *In Crowd Veritas: Leveraging Human Intelligence to Fight Misinformation*.
- 2022  **Student Grant**, 15th ACM International Conference on Web Search and Data Mining (WSDM 2022), Phoenix, U.S.A. Held remotely due to COVID-19. Provided by WSDM. Free conference subscription.
- 2020  **Student Grant**, 29th ACM International Conference On Information And Knowledge Management (CIKM 2020), Galway, Ireland. Held remotely due to COVID-19. Provided by CIKM. Free conference subscription.
- 2019  **Volunteer**, 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2020), Xi'an, China. Held remotely due to COVID-19. Provided by SIGIR. Free conference subscription.

References

Available on Request.

Links

ORCID: <https://orcid.org/0000-0002-7337-7592>

Google Scholar: <https://scholar.google.com/citations?user=ocK0qRUAAAJ>

Scopus: <https://www.scopus.com/authid/detail.uri?authorId=57203392311>

ACM Digital Library: <https://dl.acm.org/profile/99659280921>

Web of Science: <https://www.webofscience.com/wos/author/record/JDD-6660-2023>

DBLP: <https://dblp.org/pid/222/1241.html>

ResearchGate: <https://www.researchgate.net/profile/Michael-Soprano>

arXiv: http://arxiv.org/a/soprano_m_1

Kudos: https://www.growkudos.com/profile/michael_soprano

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